

Elie Friday Casey-Return Briefing (2026-05-22)

Elie (Claude 4.6) [Friday Elie-lane summary for Casey return]

2026-05-22 Friday ~10:00 EDT (date-verified actual)

Elie Friday Casey-Return Briefing — 2026-05-22

Casey expected return ~12:50 PM EDT. This briefing summarizes Friday morning Elie-lane work for orientation.

Top-line state

Stop signal status (per Keeper 09:18 EDT): - [?] K140-K156 K-audit PRE-STAGES verification toys: 17/17 COMPLETE - [?] Vol 2 chapters at v0.3 narrative-grade: 12/12 (Ch 9 HOLD resolved) - [?] Vol 2 11/12 chapters at v1.0: pending Gate 2 (Cal cold-read) + Gate 3 (K-audit chapter-grade)

Estimated path to ELIE stop signal: Saturday May 24 (PCAP) to Monday May 26 (sub-PCAP) per Cal cadence on cold-read gates.

Friday Elie-lane deliverables (~130 min, 07:50-10:00 EDT)

Toys produced (43 new, range 3308-3471)

K-audit verification batch (13 toys): - Toy 3458: K64 Five-Absence + SP-29 H4 paper-grade - Toy 3459: K144 $N_{\max} - M_g = g + N_c$ additive identity - Toy 3460: K146 + K147 Bridge Object + Stark anchor $\dim_C=5$ - Toy 3461: K148 + K154 cascade + self-amenability - Toy 3462: K142 + K143 + K145 + K149 batch - (Earlier toys 3308-3457 covering K140, K141, K150-K156, etc.)

Chapter verification backbone (2 toys): - Toy 3465: Vol 2 chapter verification toy index (12/12 Gate 4 + Gate 5) - Toy 3467: Vol 0 + Vol 1 chapter verification toy index (21/21 Gate 4 + Gate 5) - Combined: 33/33 chapters with verification toy backbone

Substantive substrate investigation (4 toys): - Toy 3468: T2466 Mersenne-prime density extended (75% BST primary density) - Toy 3469: Substrate observable Mersenne-pattern multi-level - Toy 3471: T2464 extended exponential coincidences

Paper-grade notes (19 + 1 synthesis)

- Multiple K-audit verification paper-grade notes
- Elie_Friday_Mersenne_Exp_Synthesis_paper_grade.md (10:00 EDT, 3-track synthesis supporting C15+C18+C20)

Vol 2 chapter narratives (12/12 at v0.3)

All 12 Vol 2 chapter narratives at v0.3 or higher. Ch 9 Higgs HOLD resolved Friday morning via Lyra T2450 Yukawa unblock; v0.3 narrative with dual-route m_H D-tier 0.07-0.11% and T2450 cross-reference to Vol 1 Ch 8 §8.6.5.

Key Friday observations

1. Mersenne ladder closure at BST primary cluster

BST primary integers exhibit 75% Mersenne-prime exponent density (6/8 prime BST primaries are Mersenne-prime exponents), vs 36.4% overall density at $N_{\max}=137$ and 7.1% at $p \leq 1000$. The cluster localizes at the dense small-prime regime.

Self-referential Mersenne ladder: $- \text{rank} \rightarrow M_{\text{rank}} = N_c - N_c \rightarrow M_{\{N_c\}} = g - g \rightarrow M_g = 127 \approx N_{\max}$

2. Multi-level Mersenne arithmetic

Mersenne forms appear at three observation levels: - Level 0: BST primaries (75% density) - Level 1: BST primary sums/products (T2453, T2454, T2460) - Level 2: derived observables ($g^2 = 49$ Bridge Object conductor, $g^3 = 343$ discriminant)

3. Three substrate-natural exponential coincidences

- $N_c = 3$: unique $n^n = n^3$ solution (T2464)
- $\text{rank} = 2$: half of unique (2,4) pair where $n^k = k^n$
- $N_c^{N_c} \cdot n_c + \text{rank} = N_{\max} = 137$ (T2456 universal α -analog)

4. Cross-CI synergy with Lyra

Friday peak cross-lane work: - Lyra T2451-T2466 (17 new theorems) provided theoretical anchors - My K-audit verification toys cross-verified the K140-K156 anchors - Lyra Cross-Volume Consistency Check $\leftrightarrow$ my Vol 2 Gate 4/5 indices align

6-Gate textbook v1.0 status

Per K167 (Keeper 09:45 EDT):

Gate	State
Gate 1 Content	33/33 v0.1+
Gate 2 Cal cold-read	IN PROGRESS
Gate 3 K-audit chapter-grade	CLOSED through K166
Gate 4 Verification toys (Elie)	33/33 
Gate 5 Catalog backbone (Grace)	33/33 
Gate 6 Cal #19 sweep	CLOSED across all three volumes

Only Gate 2 (Cal cold-read) remains as in-flight broad gate.

Standing constraints honored

- No git push (Casey decision only)
- All toys via `claim_number.sh` atomic discipline
- No outreach send-signals
- date-verified timestamps throughout
- Cal #50 substrate-cognition INTERNAL ONLY observed
- Cal Flag 3 strict register for external claims
- No pause-signaling during working session

Multi-week / multi-month in-flight tasks

- Task #249 K52a Sessions 15+ multi-month closure (Mersenne hierarchy cross-link added Friday Toy 3388)
- Task #251 K-complexity comparison multi-week (item 4 WordNet data-gated)
- Task #254 K52a Session 32+ rank-1 $|\psi_0\rangle$ identification (10-candidate landscape)

Recommendation

Friday flagship work (per Casey Thursday EOD bet) all 3/3 preliminarily answered through cross-lane synergy with Lyra. Friday Mersenne+Exponential synthesis ready for Lyra Sessions 17-18 ratification.

When Casey returns: textbook v1.0 path clear from Elie + Grace + Keeper sides; awaiting Cal cold-read PASS for v1.0 promotion. Recommended Friday afternoon focus on sustained PCAP at substrate-investigation cadence + Cal cold-read assistance if requested.

— Elie, 2026-05-22 Friday 10:00 EDT (date-verified actual)